(Please write your Enrollment Number)

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
(B.Tech IT) (4th Sem)

(April -May, 2024)

Subject Code: BIT-204 | Subject: Object Oriented Programming

Time : 3 Hours | Maximum Marks :60

Note: Q1. is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		(5 x 4 =20)	

	(a) In C++, the value of a comma-separated list of expressions is the value of the right-most expression. Following this, what is the output of the following program? Justify your answer.
	
	`#include <iostream>`
	`using namespace std;`
	
	`int main() {`
	`int i, j;`
	
	`j = 10;`
	`i = (j++, j+100, 999+j);`
	
	`cout << i;`
	
	`return 0;`		

	(b) Write a C++ program to find minimum of two numbers using an inline function.		
	(c) Explain the following characteristics of Java: 1) Java is secure 2) Java is portable.		
	(d) Differentiate between abstract class and interface in java through suitable program examples.		
	UNIT-I		
Q2	a) Write a c++ program to accept any 20 positive numbers and find the sum of even numbers and sum of odd numbers. b) What is typecasting? Demonstrate through a program - explicit and implicit type conversions.	(6,4)	
Q3	a) Explain any 5 characteristics of object-oriented programming. b) Write a C++ program to square an integer value using 1) Call by pointers 2) Call by reference.	(5,5)	
	UNIT-II		
Q4	a) Write a C++ operator overloading program where (++) operator increments the value by 20. b) Why is copy constructor used? Write a small C++ program to demonstrate the usage of copy constructor.	(6,4)	
Q5	a) Write a C++ program to show the order of constructor calls in Multiple Inheritance utilizing parameterized constructors. (b) Write a C++ program to demonstrate the usage of protected access specifier.	(7,3)	

	UNIT-III		
Q6	Write a C++ program that reads a text file and finds the number of characters, words and lines in a file.	**(10)**	
Q7	Explain the role of exception handling in C++ through a neat diagram. Write a C++ program to show multiple catch blocks for a single try block.	**(10)**	
	UNIT-IV		
Q8	a) Explain with small programs: 1) Packages in Java 2) Multi-threaded programing in Java	**(5,5)**	
Q9	a) Write a Java program to demonstrate multilevel inheritance utilizing super keyword to call superclass constructors.	**(10)**	